Bias Detection Tool

A machine learning pipeline for detecting and analyzing bias in datasets and AI/ML models — identifying unfair disparities across demographic groups like gender, race, and age to promote fairness and transparency in AI systems.

ACTIVE | MIT LICENSE | PYTHON 3.8+ | JUPYTER NOTEBOOK

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Agenda

A guided walkthrough of the Bias Detection Tool — from motivation to results.

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Overview

Why bias detection matters in ML

02

Features

Key capabilities of the tool

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Tech Stack

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How It Works

End-to-end ML pipeline walkthrough

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Results

Metrics, visualizations, and insights

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The Problem with Biased AI

Unchecked bias in ML models leads to discriminatory outcomes — real harm to real people. This tool provides a full end-to-end pipeline to surface and address those disparities.

Load & Preprocess

Ingest datasets and prepare features for analysis

Analyze Distributions

Examine outcomes across protected attributes

Evaluate Fairness

Quantify disparate impact across demographic groups

Visualize & Act

Surface actionable insights with intuitive plots

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Key Features

Exploratory Data Analysis

Visualize label distributions and demographic breakdowns across the full dataset

Fairness Metrics

Compute statistical parity, disparate impact, equal opportunity difference, and more

Bias-Aware Model Evaluation

Train classifiers and generate reports that surface group-level performance gaps

Sensitive Attribute Detection

Automatically identify features correlated with protected demographic attributes

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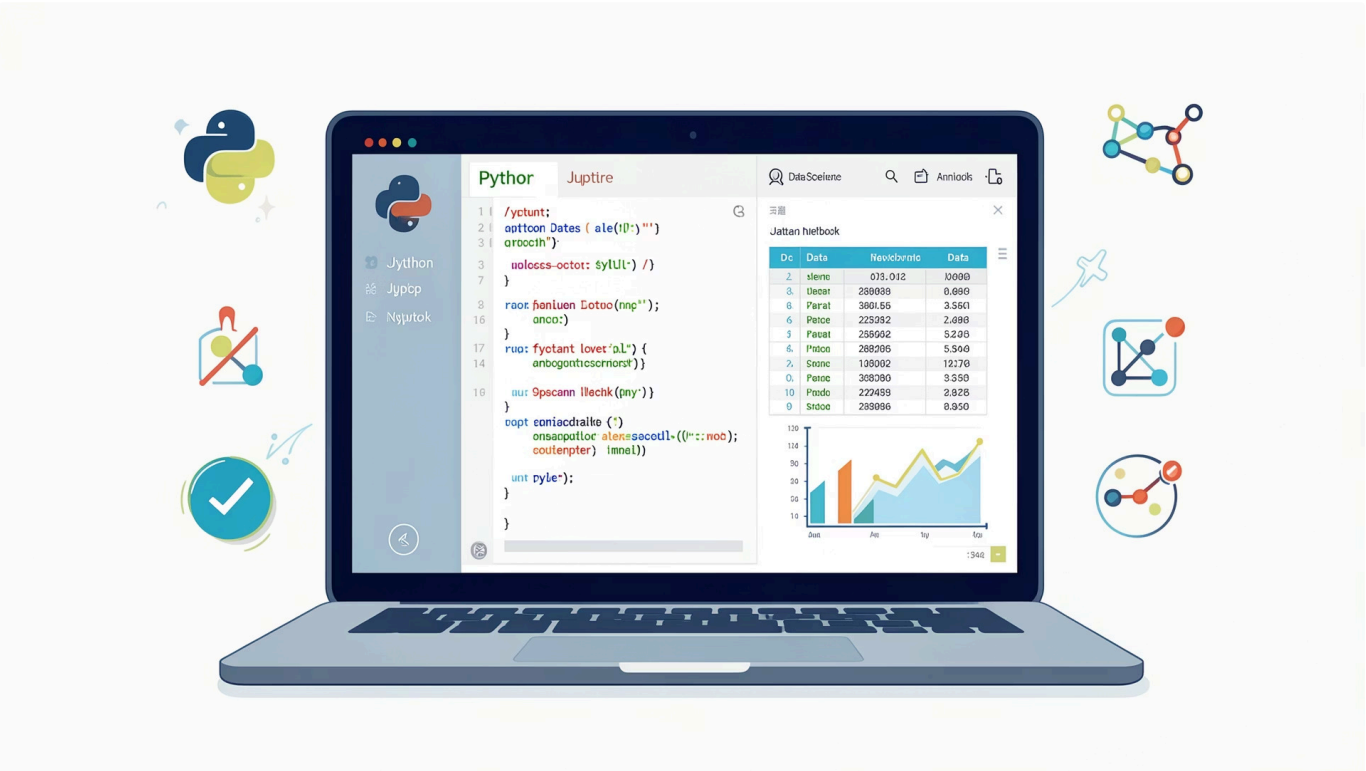
Tech Stack



Core Technologies

Python 3.8+	Core language
Jupyter Notebook	Interactive development
Pandas	Data manipulation & analysis
NumPy	Numerical computing
Scikit-learn	ML models & metrics
Matplotlib / Seaborn	Data visualization

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Project Structure

Minimal by Design

The project is intentionally lean — a single self-contained notebook that runs the full pipeline, paired with documentation. Everything you need, nothing you don't.

> The entire pipeline — from raw data to fairness metrics — lives inside `project.ipynb`. Just open and run.

```
Bias-Detection-Tool/
 │
 ├── project.ipynb   # Main pipeline notebook
 └── README.md       # Project documentation
```

All analysis, model training, fairness evaluation, and visualizations are contained in a single reproducible Jupyter Notebook — making it easy to inspect, modify, and share.

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How It Works: The Pipeline

Each stage builds on the last — raw data flows through cleaning, analysis, and modeling before fairness metrics are computed and visualized, ensuring a fully reproducible audit trail.

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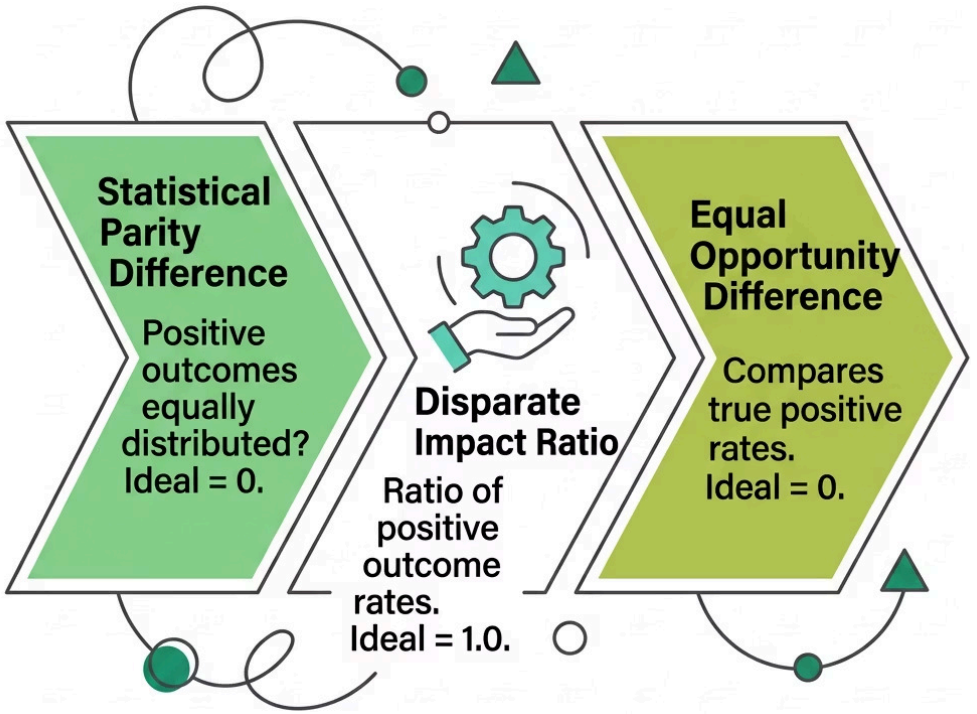
Fairness Metrics Explained

Three core metrics quantify bias across demographic groups, each measuring a different dimension of fairness:

> A model can pass one fairness criterion and fail another — evaluating all three together gives a complete picture.



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Results & Insights

The notebook delivers quantitative metrics and rich visualizations that make disparities immediately visible and actionable.

📉

Underrepresentation

Identify which demographic groups are underrepresented or systematically disadvantaged in model outcomes

🧩

Performance Breakdown

Disaggregate accuracy, precision, recall, and F1 by sensitive attribute for transparent reporting

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Before vs. After

Compare fairness metrics pre- and post-adjustment to measure the impact of mitigation strategies

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Get Started & Contribute

Quick Start

Clone the repo, install dependencies, and launch the notebook in three steps:

1. Clone: `git clone https://github.com/sonu786786/Bias-Detection-Tool`
2. Install: `pip install -r requirements.txt`
3. Run: `jupyter notebook project.ipynb`

Author & License

Built by **Sonu and Aditya**— open for collaboration and contributions from the AI fairness community.

- License: MIT — free to use, modify, and distribute
- Contributions welcome via pull requests

Star the repo and open an issue to get involved in making AI more fair.

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